

University of Asia Pacific
Department of Computer Science and Engineering
CSE 402: Operating System Lab

Lab Report: 05

Report Title: Priority Scheduling Preemptive & Non Preemptive

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1.Problem Statement:

For the given processes, perform Priority Scheduling using both Preemptive and Non-Preemptive approaches. Calculate the Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), Average Turnaround Time (Average TAT), and Average Waiting Time (Average WT). Finally, compare the Preemptive and Non-Preemptive Priority Scheduling approaches based on Average TAT and Average WT and determine which approach performs better.

Process	Arrival Time	Priority	Burst Time
P1	0 ms	3	3 ms
P2	1 ms	2	4 ms
P3	2 ms	4	6 ms
P4	3 ms	6	4 ms
P5	5 ms	10	2 ms

2.Source Code:

```
processes = [
   ["P1", 0, 3, 3],
   ["P2", 1, 2, 4],
   ["P3", 2, 4, 6],
   ["P4", 3, 6, 4],
   ["P5", 5, 10, 2]
]

n = 5

time = 0
done = []
ct = [0] * n
sequence = []

while len(done) < n:

   available = []

   for i in range(n):
      if i not in done and processes[i][1] <= time:
```

```
            available.append(i)

    if not available:
        time += 1
        continue

    x = min(available, key=lambda i: processes[i][2])

    time += processes[x][3]
    ct[x] = time

    done.append(x)
    sequence.append(processes[x][0])


tat = []
wt = []

for i in range(n):
    tat.append(ct[i] - processes[i][1])
    wt.append(tat[i] - processes[i][3])

avg_tat_non = sum(tat) / n
avg_wt_non = sum(wt) / n


print("\nNON-PREEMPTIVE PRIORITY")

print("Execution Sequence : ", " -> ".join(sequence))

print("\n{:<10}{:<10}{:<10}{:<10}".format(
    "Process", "CT", "TAT", "WT"
))



for i in range(n):
    print("{:<10}{:<10}{:<10}{:<10}".format(
        processes[i][0], ct[i], tat[i], wt[i]
```

```
    ))

print("Average TAT : {:.2f} seconds".format(avg_tat_non))
print("Average WT  : {:.2f} seconds".format(avg_wt_non))

remaining = []

for i in range(n):
    remaining.append(processes[i][3])

ct = [0] * n
time = 0
sequence = []

while sum(remaining) > 0:

    available = []

    for i in range(n):
        if processes[i][1] <= time and remaining[i] > 0:
            available.append(i)

    if not available:
        time += 1
        continue


    x = min(available, key=lambda i: processes[i][2])

    # Run for 1 second
    remaining[x] -= 1
    time += 1


    if len(sequence) == 0 or sequence[-1] != processes[x][0]:
        sequence.append(processes[x][0])

    if remaining[x] == 0:
        ct[x] = time
```

```
tat = []
wt = []

for i in range(n):
    tat.append(ct[i] - processes[i][1])
    wt.append(tat[i] - processes[i][3])

avg_tat_pre = sum(tat) / n
avg_wt_pre = sum(wt) / n


print("\nPREEMPTIVE PRIORITY")

print("Execution Sequence : ", " -> ".join(sequence))

print("\n{:<10}{:<10}{:<10}{:<10}".format(
    "Process", "CT", "TAT", "WT"
))
for i in range(n):
    print("{:<10}{:<10}{:<10}{:<10}".format(
        processes[i][0], ct[i], tat[i], wt[i]
    ))
print("Average TAT : {:.2f} seconds".format(avg_tat_pre))
print("Average WT  : {:.2f} seconds".format(avg_wt_pre))


print("\nFINAL COMPARISON")

print("\n{:<20}{:<15}{:<15}".format(
    "Method", "Average TAT", "Average WT"
))

print("{:<20}{:<15.2f}{:<15.2f}".format(
    "Non-Preemptive",
    avg_tat_non,
    avg_wt_non
))
```

```
print("{:<20}{:<15.2f}{:<15.2f}".format(
    "Preemptive",
    avg_tat_pre,
    avg_wt_pre
))

if avg_tat_non < avg_tat_pre and avg_wt_non < avg_wt_pre:
    print("Result : Non-Preemptive is BETTER")
else:
    print("Result : Preemptive is BETTER")
```

GitHub: https://github.com/Adika2003/CSE-402-Operating-System-Lab

3.Result:

```
C:\Users\adika\PycharmProjects\PythonProject1Adika\.venv\Scripts\python.exe C:\Users\adika\PycharmProjects\PythonProject1Adika\23101162_Priority_Scheduling_Lab-6.py

NON-PREEMPTIVE PRIORITY
Execution Sequence :  P1 -> P2 -> P3 -> P4 -> P5

Process   CT        TAT       WT
P1        3         3         0
P2        7         6         2
P3        13        11        5
P4        17        14        10
P5        19        14        12
Average TAT : 9.60 seconds
Average WT  : 5.80 seconds
```

```
PREEMPTIVE PRIORITY
Execution Sequence :  P1 -> P2 -> P1 -> P3 -> P4 -> P5

Process   CT        TAT       WT
P1        7         7         4
P2        5         4         0
P3        13        11        5
P4        17        14        10
P5        19        14        12
Average TAT : 10.00 seconds
Average WT  : 6.20 seconds

FINAL COMPARISON

Method              Average TAT    Average WT
Non-Preemptive      9.60           5.80
Preemptive          10.00          6.20
Result : Non-Preemptive is BETTER

Process finished with exit code 0
```

4. For this problem, which is better: Preemptive or Non-Preemptive Priority Scheduling?

Ans: For this particular problem, Non-Preemptive Priority Scheduling is better than Preemptive Priority Scheduling based on the calculated Average Turnaround Time (TAT) and Average Waiting Time (WT). In both methods, a smaller priority number represents a higher priority. In the Non-Preemptive approach, once a process starts execution, it continues until its burst time is completed, even if another process with a higher priority arrives during its execution. For the given processes, the execution sequence is P1 → P2 → P3 → P4 → P5, resulting in an Average TAT of 9.6 ms and an Average WT of 5.8 ms.

In contrast, the Preemptive Priority Scheduling approach allows a running process to be interrupted when a higher-priority process arrives. In this problem, P1 starts at 0 ms, but when P2 arrives at 1 ms with a higher priority, P1 is preempted and P2 starts executing. After P2 finishes, P1 resumes and completes. The execution sequence becomes P1 → P2 → P1 → P3 → P4 → P5. This results in an Average TAT of 10.0 ms and an Average WT of 6.2 ms.Therefore, when we compare the two approaches, the Non-Preemptive method has a lower Average TAT and a lower Average WT:

- Average TAT: 9.6 ms<10.0 ms
- Average WT: 5.8 ms<6.2 ms

This means that, for this particular set of processes, Non-Preemptive Priority Scheduling provides better overall performance. It reduces both the average time processes spend in the system and the average time they wait in the ready queue. Although Preemptive Priority Scheduling can be useful when higher-priority processes need immediate CPU access, in this specific problem the extra preemption does not improve the average waiting or turnaround time. Hence, Non-Preemptive Priority Scheduling is the better approach for this problem.

5.Discussion:

In this problem, Priority Scheduling is performed using both Non-Preemptive and Preemptive approaches. Here, a smaller priority number represents a higher priority. In the Non-Preemptive approach, once a process starts execution, it continues until completion. The execution sequence is P1 → P2 → P3 → P4 → P5**,** with an Average TAT of 9.6 ms and an Average WT of 5.8 **ms**.

In the Preemptive approach, a running process can be interrupted when a higher-priority process arrives. Here, P1 starts first, but when P2 arrives with a higher priority, P1 is preempted. The execution sequence becomes P1 → P2 → P1 → P3 → P4 → P5**.** This approach gives an Average TAT of 10.0 ms and an Average WT of 6.2 ms**.**

By comparing both approaches, the Non-Preemptive Priority Scheduling performs better for this particular problem**,** because it has lower Average TAT and Average WT. Although Preemptive Scheduling provides better response to higher-priority processes, the additional preemption does not improve the overall average performance in this case. Therefore, Non-Preemptive Priority Scheduling is the more suitable approach for the given processes**.**